AY2023/24 SEMESTER 2

MH3511 - DATA ANALYSIS WITH COMPUTER

Title: Exploratory Data Analysis of Drivers' Performance, Standings and Earnings in Formula 1 Dataset

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1. Introduction

Formula 1 (F1) is the most popular motorsport in the world today, demanding peak performances from the drivers, constructors and machines. The cars are capable of reaching 230 mph on the most difficult circuits around the world. Cars moving at such high speeds require years of experience and training from the driver to manoeuvre around the circuits. The most decorated F1 driver today is Max Verstappen who has a reported annual salary of $55 million in 2024. He has won 3 straight championships which requires him to have the most number of points at the end of the year (Fakas-Drosos, 2023).

Our report aims to use various statistical analysis including ANOVA test, fisher test, correlation coefficient, linear regression and Kruskal-Wallis test to study the relations between various factors that we determined to influence the success of a driver. Other than the performance of a driver, we will also delve into the financial aspect of the 20 drivers that make up the current F1 grid and see how their performance affects their salary. By analysing these statistical connections, we can gain valuable insights and make informed conclusions about the world of F1.

The dataset we mainly used is “Formula 1 World Championship (1950 - 2023)” from Kaggle.com. The dataset contains 14 data frames and the ones we employed are “drivers.csv”, “results.csv”, “driver_standing.csv”. We also added additional sources that we scrapped from two other webpages for more reliable insights regarding drivers’ points and salaries specific to the year of 2023, taking salary figures from RacingNews365.com and the official Formula 1 website for drivers’ standings data (refer to References for links of the 3 sources).

Based on this data set, we seek to answer the following questions:

1. Does the driver’s experience affect the average points earned per race the driver participated in in the most recent 10 years?
2. Does the pole position affect the final position of the driver?
3. Does the driver’s experience affect the driver’s salary?
4. Does the driver’s average points affect the driver’s salary?

2. Data Preparation and Analysis

We will now look into the variables used in our statistical analysis.

2.1 Summary statistics of driver's experience level and average points earned per race for each driver in the most recent 10 years

First, for F1 races that occurred in the most recent 10 years, we find the average points earned per race for each specific driver by with:

$$average\ points\ earned\ per\ race\ = \frac{total\ points\ earned\ in\ all\ races\ driver\ took\ part\ in}{total\ number\ of\ races\ driver\ took\ part\ in}$$

for the most recent 10 years in the dataset.

We define the experience level as the total number of races drivers participated in in the course of their career, which would be recorded in the vast dataset. The drivers are split into different experience_level (Low, Med or High) according to how many races they have participated in their entire career. Low: 0-99 races; Medium: 100-199 races; High: more than 200 races.

Boxplot below shows the effect of the Experience Level on Average points earned per race in the last 10 years. It only contains data of drivers who participated in at least one race in the past 10 years.

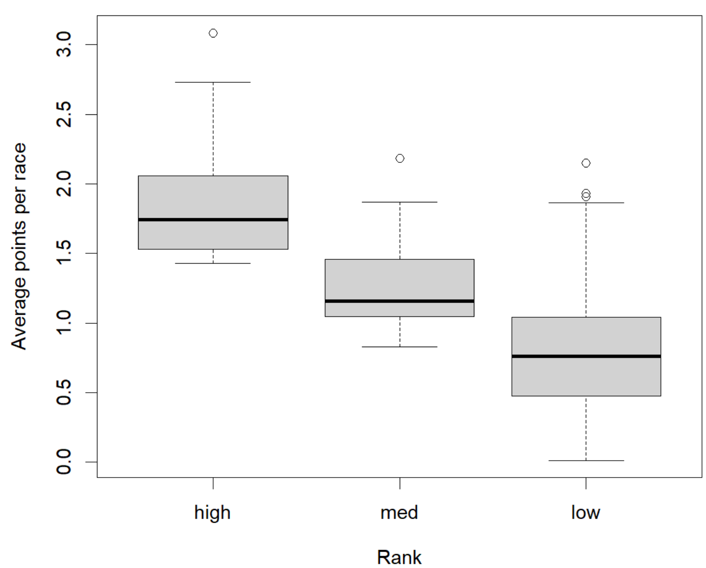

Figure 1: Boxplot of the effects of Experience Level on Average points earned per race in last 10 years before removing outliers

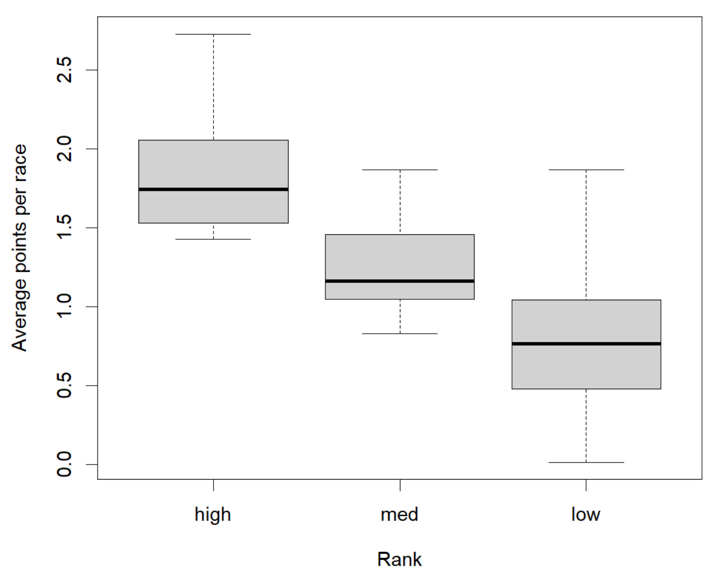

Figure 2: Boxplot of the effects of Experience Level on Average points earned per race in last 10 years after removing outliers

We observe that boxplots, after removing outliers, the interquartile range of each boxplot is similar.

2.2 Summary Statistics for Effects of starting from Pole Position on Final Position

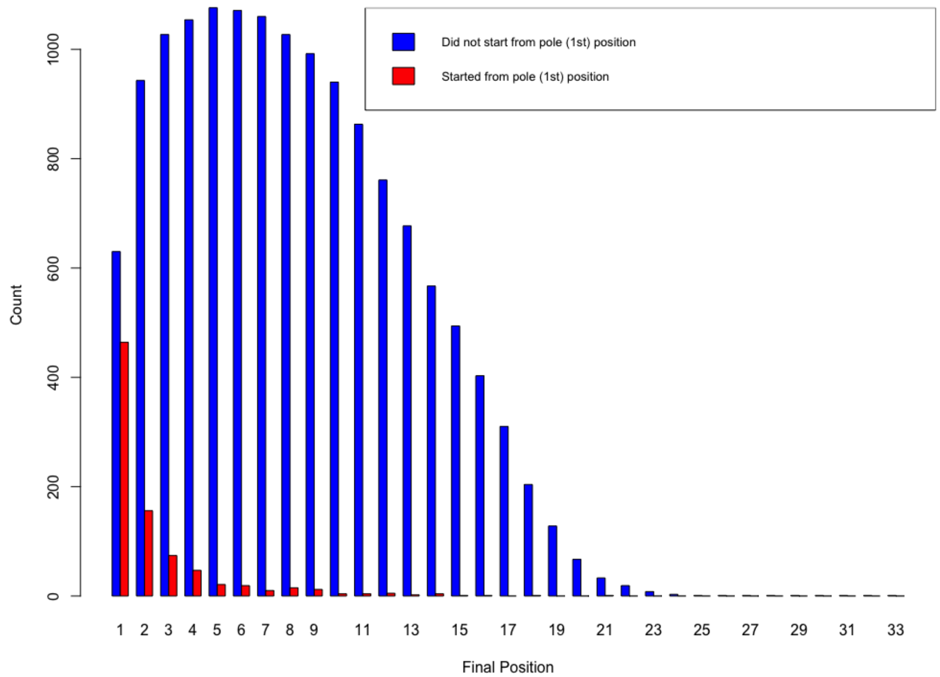

Figure 3: Effects of starting from pole position on final positions

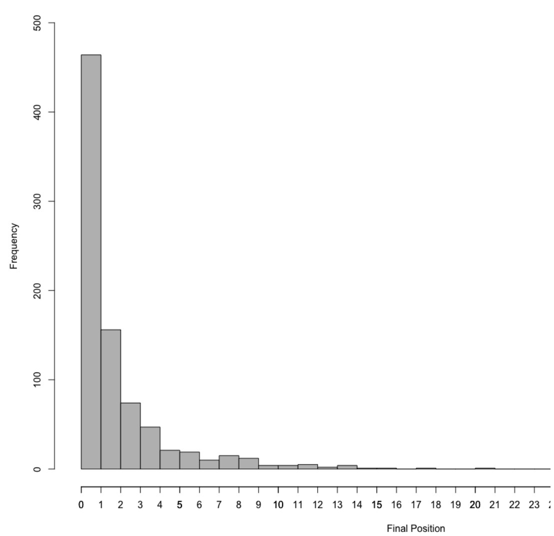

Figure 4: Histogram for final positions of drivers starting from pole position

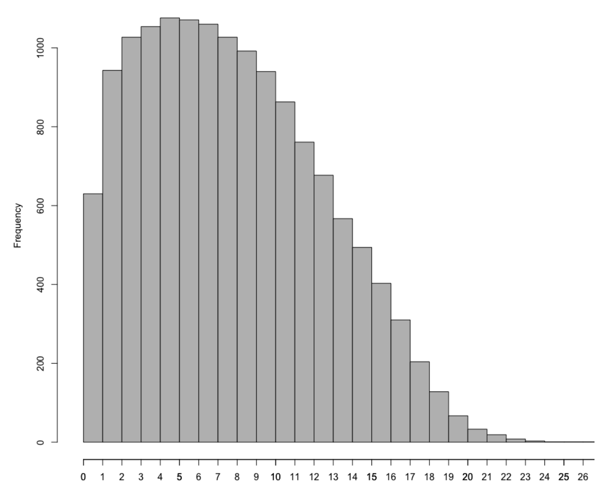

Figure 5: Histogram for final positions of drivers who did not start from pole position

This shows that the most drivers who started from pole position will end up getting the first position and those who did not start from pole position mostly obtained 5th to 6th place.

2.3 Summary statistics for drivers’ experience level and drivers’ salaries in 2023

We defined the experience of a driver as the number of races they have participated in. We took the 20 drivers that make up the current F1 grid from RacingNews365.com. The mean number of races the 20 drivers participated in is 147.65, maximum is 370 and minimum is 12.

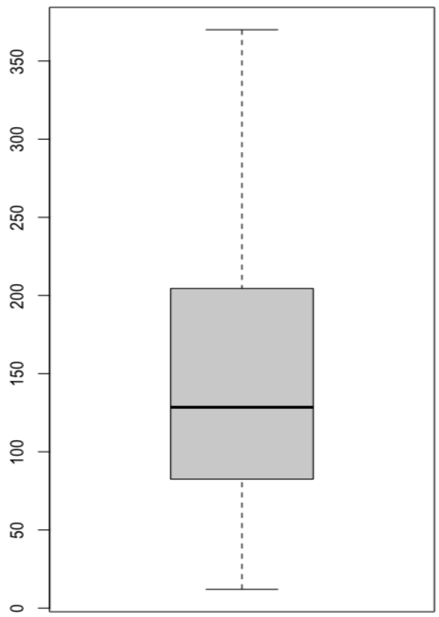

Figure 6: Boxplot of number of races participated in

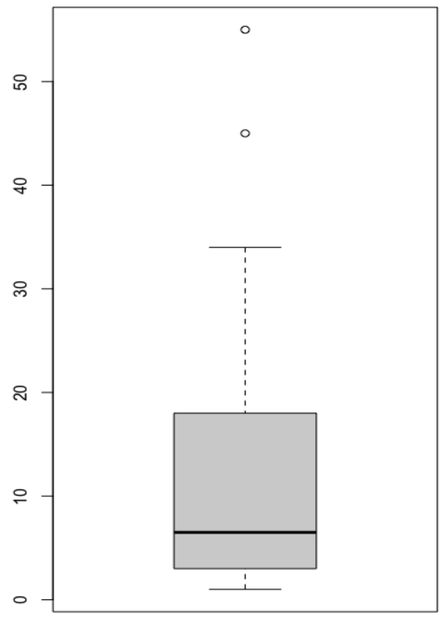

Figure 7: Boxplot of drivers’ salaries

The drivers are split into different experience levels (Low, Medium or High) according to how many races they have participated in. Low: 0-99 races; Medium: 100-199 races; High: more than 200 races.

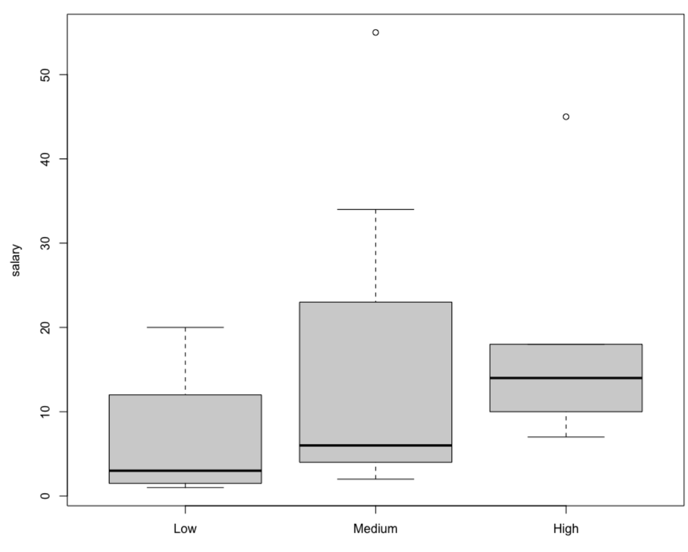

Figure 8: Boxplot of drivers’ salaries by experience level

2.4 Summary statistics for drivers' total points and drivers' salaries in 2023

After scraping, drivers' points data from formula1.com had initially one outlier and was relatively skewed from a normal distribution, as observed in the qqplot and boxplot performed on the dataset.

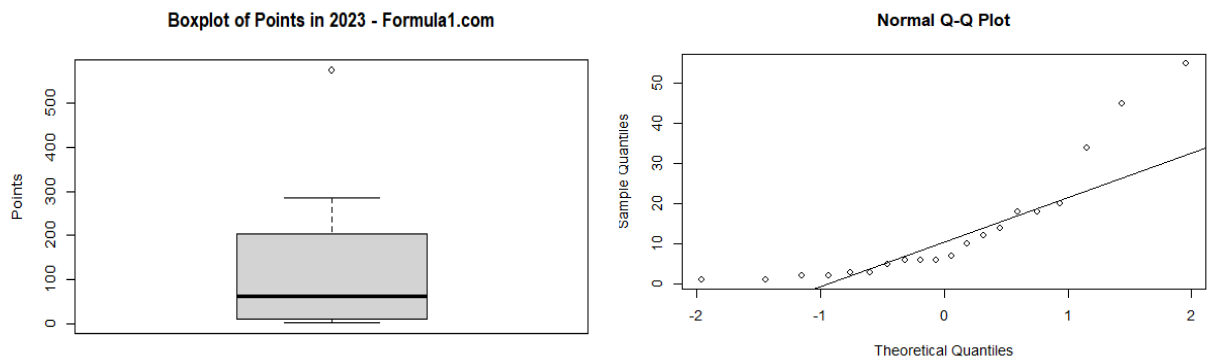

Figure 9: box plot and qq plot of drivers' points

To clean the data, the outlier was removed and logarithmic transformations were applied to the data. The following qqplot shows improvements in normality and reduced skew of the dataset.

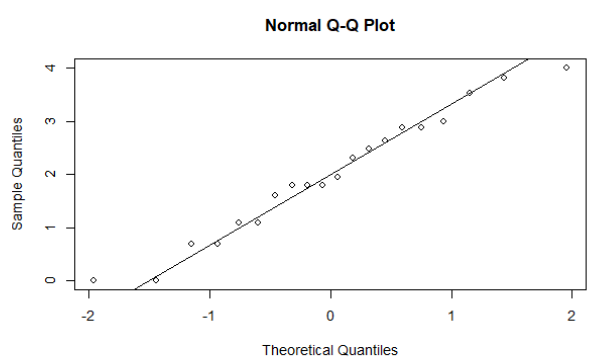

Figure 10: qq plot of log drivers' points

Upon web scraping, drivers' salaries data was observed to be skewed towards lower values and did not follow a normal distribution. 2 outliers were observed, before logarithmic transformation.

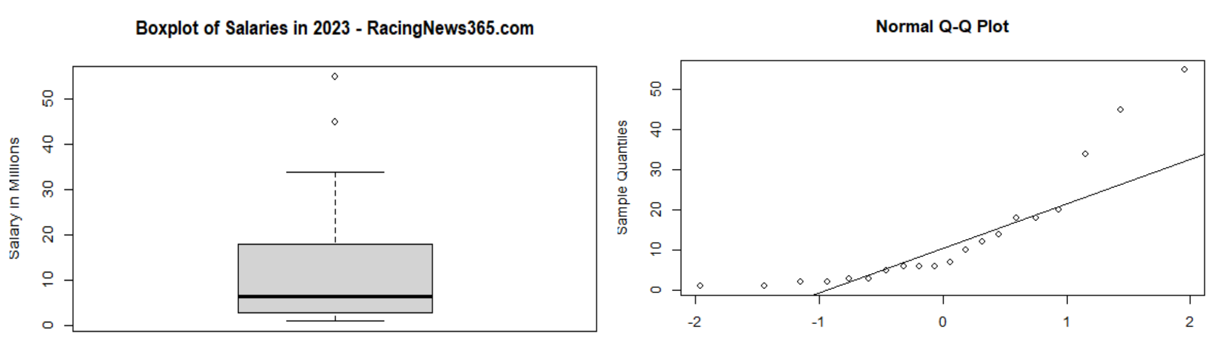

Figure 11: box plot and qq plot of drivers' salaries

After logarithmic transformation, no outliers were observed subsequently in the data. qqplot shows that the data has reduced skew and fit closer to a normal distribution than before.

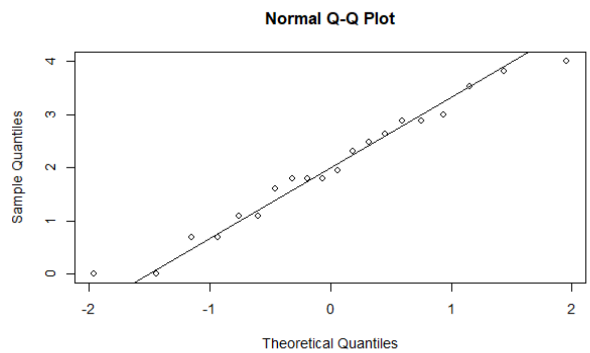

Figure 12: qq plot of log drivers' salaries

Histogram to show the distribution of drivers' points upon logarithmic transformation. Total sample size is 20 drivers with a mean of 3.531, maximum of 5.652 and minimum of 0.

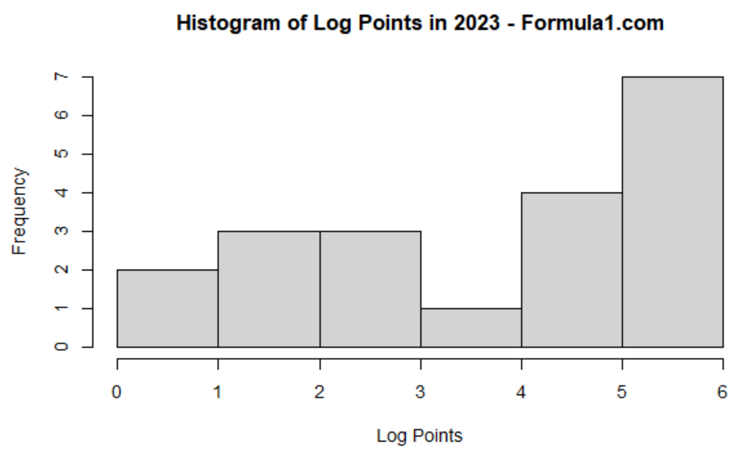

Figure 13: Histogram of log points

Histogram to show the distribution of drivers' salaries upon logarithmic transformation. Total sample size is 20 drivers with a mean of 2.003, maximum of 4.007 and minimum of 0.

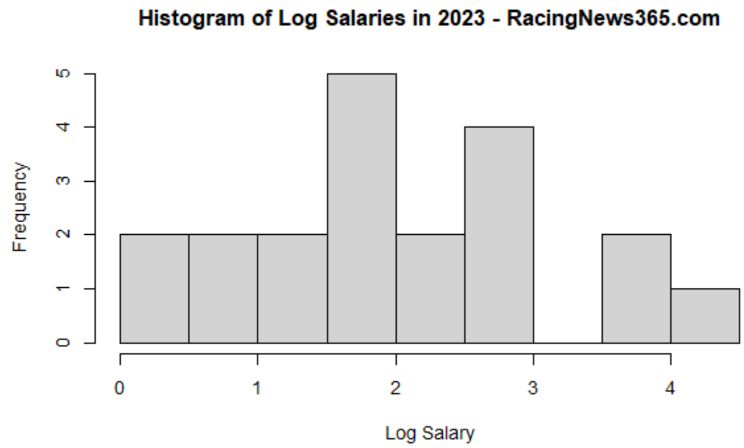

Figure 14: Histogram of log salaries

Data for this section pertains exclusively to the year 2023, which stands as the most recently completed year of races and events, offering a more up-to-date and accurate perspective on the analysis.

3. Statistical Analysis

3.1 Analysis of the relationship between drivers' experience level and average points per race in most recent 10 years

It is a common fact that "practice makes perfect", thus we will assume that the drivers with more experience will tend to have higher average points (be placed in higher ranks) per race. Therefore, we will analyse the average points per race each driver participated in the most recent 10 years of the dataset with the number of F1 races they participated in over the course of their career.

Noting that the boxplots, after removing outliers (Fig. 1), the interquartile range of each boxplot is similar, we infer that the data for each experience level (rank at "Low", "Med", "High") has similar variances.

Moreover, under the assumption that the data for each driver is independent and that it is normally distributed, we use ANOVA to find if the mean average points per race each driver participated in is affected by their different experience levels over the past 10 years. ANOVA is appropriate here when we are interested in knowing if there is an overall effect of the factor, experience level, on the outcome measure, the average points earned per race, where the data for the groups are independent, normally distributed and have similar variances.

```
> # Perform ANOVA
> anova_result <- aov(avrPointsPerRaceLast10yrs ~ factor(experience_level), data = cleaned_final)
> anova_result
Call:
   aov(formula = avrPointsPerRaceLast10yrs ~ factor(experience_level),
    data = cleaned_final)

Terms:
                factor(experience_level) Residuals
Sum of Squares                  9.280533 29.596717
Deg. of Freedom                        2        51

Residual standard error: 0.7617925
Estimated effects may be unbalanced
> summary(anova_result)
                         Df Sum Sq Mean Sq F value   Pr(>F)    
factor(experience_level)  2  9.281    4.64   7.996 0.000954 ***
Residuals                51 29.597    0.58                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> 
```

H0: mean average points per race for each race each driver of different experience levels over the past 10 years are equal.
H1: mean average points per race for each race each driver of different experience levels over the past 10 years are not equal.

The p-value (0.000954) from ANOVA test conducted on the average points per race for each driver is lower than 0.05, showing that there is sufficient evidence to reject H0, and thus the mean average points per race for each race each driver of different experience levels over the past 10 years are not equal.

We proceed with a pairwise comparison by using a pairwise t-test to determine which group ("Low", "Med", "High") is different from the other groups.

```
> pairwise.t.test(cleaned_final$avrPointsPerRaceLast10yrs, cleaned_final$experience_level, p.adjust.method = "none")

        Pairwise comparisons using t tests with pooled SD

data:  cleaned_final$avrPointsPerRaceLast10yrs and cleaned_final$experience_level

    high    low
low 0.00026 -
med 0.05678 0.08858

P value adjustment method: none
```

From the pairwise t-test, we see that high and medium experience levels (with p-value = 0.05678 > 0.05), and low and medium experience levels (with p-value = 0.08858 > 0.05) are quite similar in average points earned per race. However, low and high experience levels are very different in average points earned, with p-value = 0.00026 < 0.05.

3.2 Analysis of the relationship between pole position and final position

For a F1 race, understanding the impact of pole position on race is essential as F1-teams and drivers rely on data and analysis to develop race strategies. If pole position has a significant impact on final positions, F1-teams can adapt their strategies by prioritising their qualifying performances.

```
> # Perform Fisher's exact test with simulation
> fisher_test <- fisher.test(contingency_table, simulate.p.value = TRUE)
> fisher_test

        Fisher's Exact Test for Count Data with simulated p-value
        (based on 2000 replicates)

data:  contingency_table
p-value = 0.0004998
alternative hypothesis: two.sided
```

H0: Starting from pole position does not affect the final position.
H1: Starting from pole position does affect the final position

By performing Fisher's exact test with simulation, we are testing whether the observed association between starting from the pole position and final position is statistically significant, under the assumption that H0 is true. Since p-value is lesser than 0.05, we have sufficient evidence to reject H0 and conclude that pole position does affect the final position.

3.3 Analysis of the relationship between drivers' salaries and drivers' experience levels

When we look at the financial aspect of F1, we will assume that drivers with more experience tend to have higher salaries than drivers with lesser experience. This can also be observed from figure 8 which shows that drivers' salaries are higher as they gain more experience. Therefore, we will analyse the salaries of drivers in 2023 with differing experience levels. We deduced that the variances in salaries of the drivers with differing experience levels are not equal as seen from the boxplot in section 2.3. We concluded that ANOVA test will not be suitable as it requires the variances of each group to be equal. Therefore, we decided to utilise the Kruskal-Wallis test which is a non-parametric test.

```
> kruskal.test(currentdrivers$salary, currentdrivers$experience_level)

        Kruskal-Wallis rank sum test

data:  currentdrivers$salary and currentdrivers$experience_level
Kruskal-Wallis chi-squared = 3.7215, df = 2, p-value = 0.1556
```

H0: Median salary of drivers of different experience levels are equal.
H1: Median salary of drivers of different experience levels are not equal.

The p-value (0.1556) from the Kruskal-Wallis test conducted on the drivers' salaries shows that H0 is not rejected and thus the salaries of drivers of different experience levels may be equal. We then take a look at the correlation between drivers' salaries and their experience levels.

```
> correlation_experience_salary = cor(currentdrivers$experience, currentdrivers$salary)
> correlation_experience_salary
[1] 0.4143743
```

We observe that the correlation coefficient (0.4143743) indicates that there is a moderate level of correlation between experience levels and salaries. While the p-value shows that it is probable for the median salary to be equal, it does not mean that there is no relation between salaries and experience levels as we can see from the correlation coefficient that there is still some correlation between salaries and experience levels.

3.4 Analysis of correlation between drivers' salaries and drivers' points

Understanding the potential correlation between drivers' salaries and the points they have achieved during a specified timeframe allows a deeper understanding of how effective compensation structures are put in place for drivers to achieve their respective performances. This can prove useful in projecting drivers' salaries based on their achieved points, and understanding the general magnitude of the incentive strategies put in place to reward players
in the industry.

Linear regression is used because it is able to express the relationship between the two factors in concern as an equation, even allowing the projection of salaries with increased points. Logarithmic transformations are accounted for in the final regression equation.

```
Call:
lm(formula = merged_salariesAndPoints$logSalary ~ merged_salariesAndPoints$logPoints)

Coefficients:
                       (Intercept)  merged_salariesAndPoints$logPoints
                           0.09866                             0.50329
```

Calculating the equation for the best-fit line between the transformed salary and points data results in the following:

$$\log(y) = 0.50329 * \log(x) + 0.09866$$

Where y is the salary (in millions, USD) obtained by a driver
x is the number of points obtained by a driver

Our observations of a significant R^2 value suggests the relationship between drivers' salaries and their performance points, and how these variables are closely linked to each other. Points achieved can therefore serve as a reliable predictor of salaries pertinent to the dataset, and possibly for subsequent years as well.

4. Conclusion

In conclusion, analysing the data we have acquired provides several interesting insights into the world of Formula 1, with regards to various factors that impact drivers' performances, standings and their earnings.

Firstly, we utilised ANOVA to observe the possible correlation between drivers' experience levels and the average points they earned for each race over the past decade. Our findings found a significant relationship between the two factors, suggesting that more experienced drivers were consistent in achieving more points than other peers in the industry.

Fisher's exact test was also useful in helping us to observe that drivers who begin races in the pole position will have a significantly higher advantage compared to their competitors, according to corresponding analysis with their final positions achieved, signifying the qualifying performance of each driver.

Regarding finances, even though the Kruskal-Wallis test showed an insignificant difference in median salaries across experience levels, a moderately positive correlation suggests a possible relationship between the factors. We inferred that experience hence played somewhat of a role in determining one's earnings, even if not a major determinant.

We finally employed linear regression to examine the correlation between drivers' salaries and their points, using the year 2023 as a recent example of observation. Our analysis revealed a strong linear relationship, and that drivers' on-track performance directly impacts their financial compensation.

Thus, our study allowed us to explore the roles leading to various types of success in Formula 1, in terms of both performance and financial factors.

5. References

Fakas-Drosos, J. (2023, November 5). Why is F1 so popular? - John Fakas-Drosos - Medium. Medium. https://medium.com/@annarchitect.ddr/why-is-f1-so-popular-2b0c3d6273a9#:~:text=The%20Thrill%20of%20Speed%20in%20F1%20Racing&text=It%20is%20this%20adrenaline%2Dpumping,the%20world's%20most%20challenging%20circuits

Formula 1 World Championship (1950 - 2023). (2023, August 24). Kaggle. https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020

Results. (n.d.). Formula 1® - the Official F1® Website. https://www.formula1.com/en/results.html/2023/drivers.html

Rencken, D. (2024, February 6). F1 driver salaries 2023. RacingNews365. https://racingnews365.com/f1-driver-salaries-2023

6. Appendix

Section 2.1 preparation of data:

```
driver_standings <- read.table("C:/Users/USER/Dropbox/JOLYNNTAN/NTU STUFF/MH3511/f1archives/driver_standings.csv", sep="," , header=TRUE)
races <- read.table("C:/Users/USER/Dropbox/JOLYNNTAN/NTU STUFF/MH3511/f1archives/races.csv", sep="," , header=TRUE)

#change header position to final placing
names(driver_standings)[names(driver_standings) == "position"] <- "final placing"

#remove redundant and unnecessary columns from dataset
driver_standings$driverStandingsId <- NULL
driver_standings$positionText <- NULL
# find the races in the past 10 years
races_last_10_years <- races[races$year >= (max(races$year) - 9), ]
races_last_10_years <- races_last_10_years[,c("raceId", "year")]
# find participant count for the 55 participants
races_participantCount <- races[,c("raceId", "year")]


race_results_last_10_years <- merge(races_last_10_years, driver_standings, by = "raceId")
races_participantCount <- merge(races_participantCount, driver_standings, by = "raceId")

driverid_count <- aggregate(raceId ~ driverId, data = race_results_last_10_years , FUN = length)
# Rename the column for clarity
names(driverid_count)[2] <- "participation_count"
```

```
######################################################################################################
# count the number of races each driver participated in their entire career
participation_count <- aggregate(raceId ~ driverId, data = races_participantCount, FUN = length)
# Rename the column for clarity
names(participation_count)[2] <- "participation_count"

# count the number of races each driver participated in the most recent 10 years
participation_count_last10years <- aggregate(raceId ~ driverId, data = race_results_last_10_years, FUN = length)
# Rename the column for clarity
names(participation_count_last10years)[2] <- "participation count in last 10 years"
######################################################################################################

# Filter participant_count based on driverId from driverid_count
filtered_participant_count <- participation_count[participation_count$driverId %in% driverid_count$driverId, ]
# Add a column called "experience_level" based on participant count
filtered_participant_count$experience_level <- ifelse(filtered_participant_count$participation_count < 100, "low",
                                                      ifelse(filtered_participant_count$participation_count >= 100
                                                             & filtered_participant_count$participation_count <= 199, "med", "high"))
# Print the updated participation count data frame
print(filtered_participant_count)

race_results_points <- race_results_last_10_years[, c("driverId", "wins", "points")]
max_points <- aggregate(cbind(wins, points) ~ driverId, data = race_results_points, FUN = max)
final <- merge(merge(filtered_participant_count, max_points, by = "driverId"), participation_count_last10years, by = "driverId")
final$avrPointsPerRaceLast10yrs <- final$points / final$`participation count in last 10 years`
experience_count <- table(final$experience_level)
```

Section 2.1 plotting of boxplot:

```
new_order <- c("high", "med", "low")  # Change this to your desired order
# Reorder the factor levels in the data
final$experience_level <- factor(final$experience_level, levels = new_order)

#average points per race (in last 10 years) against experience level (remove outlier)
boxplot(final$avrPointsPerRaceLast10yrs ~ final$experience_level, col="light gray",
        ylab = "Average points per race", xlab="Rank",
        main = "Effects of Experience Level on Average points earned per race in last 10 years",
        outline = FALSE)
```

Section 2.2: Preparation of data

```r
library(dplyr)

# Read data
pole <- read.csv("/Users/pp16/Downloads/archive/results.csv")
drivers <- read.csv("/Users/pp16/Downloads/archive/drivers.csv")

# Merge data
positions <- merge(pole, drivers, by = "driverId") %>%
  select(driverId, forename, surname, raceId, grid, position)

# Create a variable indicating whether the driver started from pole position
positions <- positions %>%
  mutate(started_from_pole = ifelse(grid == 1, "Yes", "No"))

# Sort the dataframe by the 'position' column, with '1' values grouped together
positions <- positions %>%
  arrange(position != 1, position)

# If you want to keep the original order of the dataframe intact for other columns, you can add a secondary sort by row number
positions <- positions %>%
  arrange(position != 1, position, row_number())

positions

# Create a contingency table
contingency_table <- table(positions$started_from_pole, positions$position)
contingency_table
```

Section 2.2: Plotting of boxplot and histograms

```
# Plot observed frequencies or proportions for each category
barplot(contingency_table_sorted,
        beside = TRUE,
        col = c("blue", "red"),
        xlab = "Final Position",  # Label x-axis
        ylab = "Count",            # Label y-axis
        main = "Effect of Starting from Pole Position on Final Positions")  # Add title

# Adjust the size of the legend
legend("topright",
       legend = c("Did not start from pole (1st) position", "Started from pole (1st) position"),
       fill = c("blue", "red"),
       cex = 0.8)  # Smaller size, adjust as needed

# Histogram for drivers who started from the pole position
hist(positions$position[positions$started_from_pole == "Yes"],
     main = "Histogram of Final Positions for drivers who started from Pole position",
     xlab = "Final Position",
     ylab = "Frequency",
     col = "grey",
     breaks = 0:33,  # Specify breaks from 0 to 33
     xlim = c(0, 34),  # Set the x-axis limits
    ylim = c(0,500))

# Add custom labels to the x-axis
axis(1, at = 0:33, labels = 0:33)

# Histogram for drivers who didn't start from the pole position
hist(positions$position[positions$started_from_pole == "No"],
     main = "Histogram of Final Positions (Did not Start from Pole)",
     xlab = "Final Position",
     ylab = "Frequency",
     col = "grey",
     breaks = 0:33,  # Specify breaks from 0 to 33
     xlim = c(0, 34),  # Set the x-axis limits
     ylim = c(0,1100))

# Add custom labels to the x-axis
axis(1, at = 0:33, labels = 0:33)
```

Section 2.3:

```
#filtering out the current 20 drivers
drivers = read.csv("drivers.csv")
currentdrivers = drivers[drivers$driverRef == "hamilton" | drivers$driverRef == "max_verstappen" | drivers$driv
salary = c(45, 18, 6, 2, 14, 7, 10, 5, 55, 12, 6, 3, 34, 20, 18, 3, 1, 2, 6, 1)
currentdrivers = data.frame(currentdrivers, salary)
currentdrivers = currentdrivers[order(currentdrivers$driverId),]
boxplot(currentdrivers$salary)

#tabulating total number of races each driver participated in
results = read.csv("results.csv")
race_count = table(results$driverId)
race_count_df = data.frame(race_count)
currentdrivers_id = currentdrivers$driverId
experience = subset(race_count_df, Var1 %in% currentdrivers_id)$Freq
currentdrivers = data.frame(currentdrivers,experience)
boxplot(currentdrivers$experience)

#splitting drivers into various experience levels based on number of races participated
cutpoints = c(0,100,200,Inf)
category_labels <- c("Low", "Medium", "High")
currentdrivers$experience_level = cut(currentdrivers$experience, breaks = cutpoints, labels = category_labels)
boxplot(currentdrivers$logsalary~factor(currentdrivers$experience_level))
```

2.4 Preparation of Data and generation of figures

```r
# Import "C:\Users\garre\Downloads\f1_driver_salaries.csv"
salaries <- read.csv("C:/Users/garre/Downloads/f1_driver_salaries.csv")
head(salaries)
# split first column of drivers into first name and last name, in salaries database
install.packages("tidyr")
library(tidyr)
# Split the 'Name' column into 'First Name' and 'Last Name'
salaries <- salaries %>%
  separate(Name, into = c("First Name", "Last Name"), sep = " ")
# Create vectors for names and points
names <- c("M. Verstappen", "S. Perez", "L. Hamilton", "F. Alonso", "C. Leclerc",
           "L. Norris", "C. Sainz", "G. Russell", "O. Piastri", "L. Stroll",
           "P. Gasly", "E. Ocon", "A. Albon", "Y. Tsunoda", "V. Bottas",
           "N. Hulkenberg", "D. Ricciardo", "Z. Guanyu", "K. Magnussen",
           "L. Lawson", "L. Sargeant")
points <- c(575, 285, 234, 206, 206, 205, 200, 175, 97, 74,
            62, 58, 27, 17, 10, 9, 6, 6, 3, 2, 1)
df
# split the 'Name' column into 'First Name' and 'Last Name'
df <- df %>%
  separate(Name, into = c("First Name", "Last Name"), sep = " ")
df
points <- df



salaries
points
# merge last name
merged_df <- merge(df, salaries, by.x = "Last Name", by.y = "Last Name")
merged_salariesAndPoints <- merged_df
merged_salariesAndPoints$logSalary <- log(merged_salariesAndPoints$Salary.in.Millions)
merged_salariesAndPoints$logPoints <- log(merged_salariesAndPoints$Points)
merged_salariesAndPoints
# isolate logSalary and logPoints
LogSalariesPointsOnly <- merged_salariesAndPoints[,c("logSalary", "logPoints")]
LogSalariesPointsOnly
```

```r
# boxplots and qqlines of salary and points
boxplot(merged_salariesAndPoints$Salary.in.Millions, main = "Boxplot of Salary in Millions", ylab = "Salary in Millions")
qqnorm(merged_salariesAndPoints$Salary.in.Millions)
qqline(merged_salariesAndPoints$Salary.in.Millions)
# points
boxplot(merged_salariesAndPoints$Points, main = "Boxplot of Points", ylab = "Points")
qqnorm(merged_salariesAndPoints$Points)
qqline(merged_salariesAndPoints$Points)
# histograms of logSalary and logPoints
hist(merged_salariesAndPoints$logSalary, main = "Histogram of Log Salary in Millions", xlab = "Log Salary in Millions")
hist(merged_salariesAndPoints$logPoints, main = "Histogram of Log Points", xlab = "Log Points")
```

Section 3.1:

```
# remove outliers
Q1 <- quantile(final$avrPointsPerRaceLast10yrs, 0.25)
Q3 <- quantile(final$avrPointsPerRaceLast10yrs, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR
outliers <- final$avrPointsPerRaceLast10yrs < lower_bound | final$avrPointsPerRaceLast10yrs > upper_bound
cleaned_final <- final[!outliers, ]
cleaned_final

# Perform ANOVA
anova_result <- aov(avrPointsPerRaceLast10yrs ~ factor(experience_level), data = cleaned_final)
anova_result
summary(anova_result)
# Pairwise t test
pairwise.t.test(cleaned_final$avrPointsPerRaceLast10yrs, cleaned_final$experience_level, p.adjust.method = "none")
```

3.4 Calculation of linear regression between points and salaries

```
# calculate correlation
cor(merged_salariesAndPoints$logSalary, merged_salariesAndPoints$logPoints)
# calculate the linear regression
lm(merged_salariesAndPoints$logSalary ~ merged_salariesAndPoints$logPoints)
```